

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

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COURSE CODE: ITA0612

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?
(5 Marks)
2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?
(5 Marks)
3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.
(5 Marks)
4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:
(5 Marks)
 - a. Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
 - b. Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
 - (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
 - (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
 - (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
 - (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
 - (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
 - (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

- i) Identify the epoch at which the model begins to overfit.
- ii) Suggest two effective techniques to reduce overfitting in this model.
- iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

- (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
- (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
- (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
- (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
- (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
- (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
- (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
- (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

- i) Identify the epoch at which the model starts overfitting.
- ii) Explain why validation accuracy decreases despite training accuracy increasing.
- iii) Suggest two methods to improve validation performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Application Based Questions

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Machine Learning

① Given.

There are three boxes:

Box A : 6 Red , 4 Blue , 2 Green = 12 Total

Box B : 5 Red , 7 Blue , 3 Green = 15 Total

Box C : 8 Red , 2 Blue , 5 Green = 15 Total

Box B is twice as likely to be selected as A or C.

Let, Probability of selecting A = x

Since B is twice as likely

$$P(A) = x, \quad P(B) = 2x, \quad P(C) = x$$

total probability is 1 :

$$x + 2x + x = 1$$

$$4x = 1$$

$$x = \frac{1}{4}$$

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{4}$$

Probability of getting a blue ball (B_i)

$$P(B_i/A) = \frac{4}{12} = \frac{1}{3}$$

$$P(B_i/B) = \frac{7}{15}$$

$$P(B_1|C) = \frac{2}{15}$$

Apply Baye's theorem,

$$P(C|B_1) = \frac{P(C) \cdot P(B_1|C)}{P(A) \cdot P(B_1|A) + P(B) \cdot P(B_1|B) + P(C) \cdot P(B_1|C)}$$

$$= \frac{\frac{1}{4} + \frac{2}{15}}{\frac{1}{4} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{7}{15} + \frac{1}{4} \cdot \frac{2}{15}}$$

$$= \frac{\frac{1}{30}}{\frac{1}{12} + \frac{7}{30} + \frac{1}{30}}$$

$$= \frac{\frac{2}{60}}{\frac{21}{60}}$$

$$P(C|B_1) = \frac{2}{21}$$

∴ The Probability that the selected box was is approximately 9.52%.

Given,

Section A : 5 Red, 3 Blue, 2 Black = 10 Total

Section B : 4 Red, 6 Blue, 5 Black = 15 Total

Section C : 7 Red, 2 Blue, 6 Black = 15 Total.

Section B is twice as likely to be selected as A or C.

$$P(A) = x, \quad P(B) = 2x, \quad P(C) = x$$

total probability = 1

$$x + 2x + x = 1$$

$$4x = 1$$

$$x = \frac{1}{4}$$

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{4}$$

Probability of selecting a blue car. (B_1)

$$P(B_1/A) = \frac{3}{10}$$

$$P(B_1/B) = \frac{2}{5}$$

$$P(B_1/C) = \frac{2}{15}$$

Apply Baye's Theorem,

$$P(C/B_1) = \frac{P(C) \cdot P(B_1/C)}{P(A) \cdot P(B_1/A) + P(B) \cdot P(B_1/B) + P(C) \cdot P(B_1/C)}$$

$$= \frac{\frac{1}{4} \cdot \frac{2}{15}}{\frac{1}{4} \cdot \frac{3}{10} + \frac{1}{2} \cdot \frac{2}{5} + \frac{1}{4} \cdot \frac{2}{15}}$$

$$= \frac{\frac{4}{120}}{\frac{9}{120} + \frac{24}{120} + \frac{4}{120}}$$

$$= \frac{4}{37}$$

$$= \frac{4}{37}$$

$$P(C/B_1) = 10.81\%$$

Probability that the blue car came from section C is $\frac{4}{37}$, 10.81% //

3) Breast Cancer classification

Given,

Malignant samples = 1800

Benign samples = 1800

Total samples:

$$1800 + 1800 = 3600$$

20% data is used for testing

$$3600 \times 20\% = 720$$

test dataset contains 720 samples

Given,

$$TP = 210$$

$$TN = 190$$

$$FP = 30$$

$$FN = ?$$

false negatives

$$TP + TN + FP + FN = \text{Total Test samples}$$

$$210 + 190 + 30 + FN = 720$$

$$430 + FN = 720$$

$$FN = 290$$

	Predicted malignant	Predicted Benign
Actually Malignant	TP = 210	FN = 290
Actually Benign	FP = 30	TN = 190

Accuracy:

$$A = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{210 + 190}{720}$$

$$= \frac{400}{720}$$

$$\text{Accuracy} = 55.56\%$$

Precision

$$P = \frac{TP}{TP + FP}$$

$$= \frac{210}{210 + 30}$$

$$= \frac{210}{240}$$

$$\text{Precision} = 87.5\%$$

Sensitivity / Recall

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{210}{210 + 290}$$

$$= \frac{210}{500}$$

$$\text{Sensitivity} = 42\%$$

Specificity:-

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$= \frac{190}{190 + 30}$$

$$\text{Specificity} = 86.36\%$$

4. Model Performance Analysis

Given.

Epoch	Training Loss	Validation
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

i) Epoch at which overfitting begins

- From Epoch 1 to 3, both Training and Validation losses decrease.
- At Epoch 3, Validation loss reaches its minimum value of 2.0.
- From Epoch 4 onward, training loss continues to decrease, but validation loss starts increasing.

Overfitting begins at Epoch 4

ii) Two techniques to reduce overfitting.

i) Dropout :-

Randomly disables some neurons during training, preventing the model from depending too much on particular features.

(4)
i) Early stopping:

Stop training when the validation loss starts increasing. Here, training can be stopped around Epoch 3, where validation loss is minimum.

ii) Effect of overfitting on Generalization

Overfitting occurs when the model learns the training data too closely, including its noise & specific patterns.

decreasing

2.8 $\rightarrow$ 1.3

increasing

2.0 $\rightarrow$ 3.2

overfitting starts at Epoch 4

5

Epoch	Training Accuracy	Validation Accuracy
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

i) Epoch at which overfitting starts
up to Epoch 5, both training and validation accuracy
generally increase

At Epoch 5:

- Training Accuracy = 80%
- Validation Accuracy = 73%

Epoch 6 onward:

- Increases : 80% - 85% - 88% - 92%
- decreases : 73% - 72% - 70% - 68%

overfitting starts at Epoch 6

ii) The model memorizing the training data instead of learning general patterns.

Training accuracy : 80% $\rightarrow$ 92%

Validation accuracy : 73% $\rightarrow$ 68%

(iii) a) Early stopping:

stop training validation accuracy, training could be stopped around Epoch 5. (73%) height

b) Dropout:

use dropout or L1/L2 regularization to prevent & reduce overfitting.